MIL-E-22200/3F
19 September 1986
SUPERSEDING
MIL-E-22200/3E
22 June 1978
(See 6.6)

MILITARY SPECIFICATION

ELECTRODES, WELDING, COVERED: NICKEL BASE ALLOY; AND COBALT BASE ALLOY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers nickel, nickel alloy, and chromium-cobalt alloy covered electrodes for shielded metal-arc welding with direct current (dc) reverse polarity.

1.2 Classification. Electrodes shall be furnished in the types, classes and sizes shown in table I as specified (see 6.2.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N3944 FSC 3439
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

TABLE I. Classification.

Type[1/]	Class	Weld position	Size (inch)
Group I [2/]			
MIL-1N12	1	All	3/32, 1/8
	2	Horizontal fillets and flat	5/32, 3/16
MIL-3N12(ENiCrFe-1)	1	All	3/32, 1/8
	2	Horizontal fillets and flat	5/32, 3/16
MIL-3N12H	1	All	3/32, 1/8
	2	Horizontal fillets and flat	5/32, 3/16
MIL-4N1A(ENiCrFe-2)	1	All	5/64, 3/32, 1/8
	2	Horizontal fillets and flat	5/32, 3/16
MIL-4N1AH	1	All	5/64, 3/32, 1/8
	2	Horizontal fillets and flat	5/32, 3/16
MIL-4N11(ENi-1)	1	All	3/32, 1/8
	2	Horizontal fillets and flat	5/32, 3/16
MIL-8N12(ENiCrFe-3)	1	All	5/64, 3/32, 1/8
	2	Horizontal fillets and flat	5/32, 3/16
MIL-8N12H	1	All	5/64, 3/32, 1/8
	2	Horizontal fillets and flat	5/32, 3/16
MIL-9N10(ENiCu-7)	1	All	5/64, 3/32, 1/8
	2	Horizontal fillets and flat	5/32, 3/16, 1/4
Group II [2/]			
MIL-3N1B(ENiMo-1)	2	Horizontal fillets and flat	3/32, 1/8, 5/32, 3/16, 1/4
MIL-3N1C(ENiMo-2)	2	Horizontal fillets and flat	3/32, 1/8, 5/32, 3/16, 1/4
MIL-3N1L	2	Horizontal fillets and flat	5/64, 3/32, 1/8, 5/32, 3/16, 1/4
MIL-3N1N	2	Horizontal fillets and flat	5/64, 3/32, 1/8, 5/32, 3/16, 1/4
MIL-4N1W(ENiMo-3)	2	Horizontal fillets and flat	3/32, 1/8, 5/32, 3/16, 1/4

1/ Identification in parentheses following MIL-types is AWS-ASTM classification.
2/ Group identification is intended to group types for weld test purposes (see tables VII and VIII).

1.2.1 Part number. Part numbers for items described in this specification will be formulated as specified in 6.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.

MILITARY

MIL-W-10430 - Welding Rods and Electrodes; Packaging of.
MIL-E-22200 - Electrodes, Welding, Covered; General Specification for.
MIL-S-22698 - Steel Plate and Shapes, Weldable Ordinary Strength and Higher Strength: Hull Structural.
MIL-N-23228 - Nickel-Chromium-Iron Alloy Plate, Sheet and Strip, Air Melted or Vacuum Remelted.

STANDARDS

FEDERAL

FED-STD-151 - Metals; Test Methods.

MILITARY

MIL-STD-271 - Nondestructive Testing Requirements for Metals.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0900-LP-003-9000 - Radiographic Standards for Production and Repair Welds.

DEPARTMENT OF LABOR

Code of Federal Regulations, Title 29
Part 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 131 - Standard Specification for Structural Steel for Ships.
A 285 - Standard Specification for Pressure Vessel Plates, Carbon Steel, Low- and Intermediate-Tensile Strength. (DoD adopted)
B 127 - Standard Specification for Nickel-Copper Alloy (UNS N04400) Plate, Sheet, and Strip. (DoD adopted)
B 162 - Standard Specification for Nickel Plate, Sheet and Strip. (DoD adopted)
B 166 - Standard Specification for Nickel-Chromium-Iron Alloys (UNS N06600 and N06690) Rod and Bar, and Wire. (DoD adopted)
B 168 - Standard Specification for Nickel-Chromium-Iron Alloys (UNS N06600 and N06690) Plate, Sheet, and Strip. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

B4.0 - Standard Methods for Mechanical Testing of Welds. (DoD adopted)
Z49.1 - Safety in Welding and Cutting.

(Application for copies should be addressed to American Welding Society, Inc., 550 NW LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Electrodes furnished under this specification shall be in accordance with MIL-E-22200 and as specified herein.

3.2 Coverings.

3.2.1 Composition. The chemical composition is optional with the manufacturer. The total water content of the coverings shall not exceed the value of 0.8 percent by weight maximum.

3.3 Mechanical properties. The mechanical properties of the deposited weld metal shall be as specified in table II.

TABLE II. Mechanical properties of deposited weld metal.

Electrode type	Tensile strength 1/ lb/in^2 (minimum)	Elongation 4X diameter percent (minimum)	Bend test 2/	Hardness 3/ Brinell (minimum)
MIL-1N12	110,000	30	X	---
MIL-3N12	80,000	30	X	---
MIL-3N12H	80,000	30	X	---
MIL-4N1A	80,000	30	X	---
MIL-4N1AH	80,000	30	X	---
MIL-4N11	60,000	20	X	---
MIL-4N12	80,000	20	X	---
MIL-8N12	80,000	30	X	---
MIL-8N12H	80,000	30	X	---
MIL-9N10	4/ 70,000	30	X	---
MIL-3N1B	80,000	10	---	165
MIL-3N1C	80,000	10	---	176
MIL-3N1L	80,000	10	---	184
MIL-3N1N	80,000	10	---	165
MIL-4N1W	80,000	10	---	165

1/ As-welded properties, except electrode type MIL-9N10.
2/ See 3.3.1.
3/ 3000 kilogram (kg), 10 millimeter (mm) ball.
4/ Properties for as-welded condition and for as-welded plus stress relieved condition.

3.3.1 Bend test fissures. Tested bend specimens shall be visually examined on the convex surface by the unaided eye, corrected if necessary to read J1 letters on the standard Jaeger's Test type chart for near vision, or equivalent type test. The specimens shall not exhibit evidence of fissuring in excess of that specified in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

TABLE III. Permissible fissures-bend tests (maximum no. (n) and length (l)).1/2/

Type	Electrode diameters (inches)											
	5/64		3/32		1/8		5/32		3/16		1/4	
	l	n	l	n	l	n	l	n	l	n	l	n
MIL-4N11	1/8	3	1/8	3	1/8	3	1/8	4	1/8	4	1/8	4
MIL-4N12	1/8	3	1/8	3	1/8	3	1/8	4	1/8	4	1/8	4
MIL-1N12, 3N12, 3N12H, 4N1A, 4N1AH	3/32	3	3/32	3	3/32	3	3/32	3	3/32	3	3/32	3
MIL-9N10	3/32	3	3/32	3	3/32	3	3/32	4	3/32	4	3/32	4
MIL-8N12	3/32	2	3/32	2	3/32	2	3/32	2	3/32	2	3/32	2
MIL-8N12H	3/32	2	3/32	2	3/32	2	3/32	2	3/32	2	3/32	2

1/ Fissures 1/64 inch and less shall not be counted.
2/ The corners of specimens shall not be considered when evaluating the results.

3.4 Chemical composition of deposited weld metal. Chemical composition of deposited weld metal shall be as specified in table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

TABLE IV. Chemical composition (percent) of deposited weld metal.3/4/

Type	Percentage (maximum unless otherwise indicated)															
	Nickel	Copper	Manganese	Iron	Silicon	Carbon	Sulfur	Chromium	Titanium	Aluminum	Columbium1/	Cobalt	Tantalum	Molybdenum	Phosphorus	Other
MIL-1N12	Balance2/	0.50	1.0	7.0	0.75	0.10	0.02	20.0/23.0	—	—	3.15/4.15	—	—	8.0/10.0	0.030	0.50
MIL-3N12	Balance	0.50	3.5	11.0	0.75	0.10	0.015	13.0/17.0	—	—	1.5/4.0	0.12	0.40	—	0.030	0.50
MIL-3N12H	Balance	0.50	3.5	11.0.	0.75	0.030/0.10	0.015	13.0/17.0	—	—	1.5/4.0	0.12	0.40	—	0.030	0.50
MIL-4N1A	Balance2/	0.50	3.5	12.0	0.75	0.10	0.02	13.0/17.0	—	—	0.50/3.0	0.12	0.30	0.5/2.5	0.030	0.50
MIL-4N1AH	Balance2/	0.50	3.5	12.0	0.75	0.030/0.10	0.02	13.0/17.0	—	—	0.50/3.0	0.12	0.30	0.5/2.5	0.030	0.50
MIL-4N11	92 Minimum	0.25	0.75	0.75	1.25	0.10	0.02	—	1.0/4.0	1.0	—	—	—	—	0.030	0.50
MIL-8N12	Balance2/	0.50	5.0/9.5	10.0	1.0	0.10	0.015	13.0/17.0	1.0	—	1.0/2.5	0.12	0.25	—	—	0.50
MIL-8N12H	Balance2/	0.50	5.0/9.5	10.0	1.0	0.030/0.10	0.015	13.0/17.0	1.0	—	1.0/2.5	0.12	0.25	—	0.030	0.50
MIL-9N10	62/68	Balance	4.0	2.5	1.0	0.15	0.015	—	1.0	0.75	—	—	—	—	0.030	0.50
Type	Nickel	Vanadium	Manganese	Iron	Silicon	Carbon	Sulfur	Chromium	Titanium	Tungsten	Columbium	Cobalt	Tantalum	Molybdenum	Phosphorus	Other
MIL-3N1B	Balance	0.60	1.0	4.0/7.0	1.0	0.12	0.030	1.0	—	—	—	2.50	—	26.0/30.0	0.030	—
MIL-3N1C	Balance	0.35	1.0	4.0/7.0	1.0	0.12	0.030	14.50/16.50	—	3.00/4.50	—	2.50	—	15.0/17.0	0.030	—
MIL-3N1L	9.0/11.0	—	0.5/2.0	3.0	1.0	0.15	0.030	19.0/21.0	—	14.0/16.0	—	Balance	—	—	0.030	—
MIL-3N1N	19.0/21.0	—	1.00/2.50	Balance	1.0	0.15	0.030	20.0/22.5	—	2.0/3.0	0.75/1.25	18.5/21.0	—	2.5/3.5	0.030	—
MIL-4N1W	Balance	0.60	1.0	4.0/7.0	1.0	0.12	0.030	2.50/5.50	—	—	—	2.50	—	23.0/27.0	0.030	—

1/ Includes incidental tantalum when not specified separately.
2/ Includes incidental cobalt when not specified separately.
3/ The weld metal shall be analyzed for the specific elements shown in this table. In the course of this work, if the presence of other elements is indicated, the amount of those elements shall be determined to ensure that their total does not exceed the limit specified for "Other elements total" in the last column of the table.
4/ Single values are maximum, except where otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

3.5 Nominal lengths. Nominal lengths of electrodes shall be in accordance with table V.

TABLE V. Nominal lengths.

Sizes (inch)	Nominal electrode lengths (inches)
5/64	9
3/32	9 or 12
1/8	1/, 2/ 9 or 14
5/32, 3/16, 1/4	14

1/ Unless otherwise specified (see 6.2.1), the longer length shall be furnished.
2/ When specified (see 6.2.1) MIL-8N12 and MIL-8N12H, 1/8-inch size may be furnished in 12-inch lengths.

3.6 Electrode identification. Marking, size, legibility and stability shall be in accordance with the requirements for electrode identification of MIL-E-22200.

3.6.1 Lot identification. When specified (see 6.2.1), each electrode shall be marked with lot identification in accordance with MIL-E-22200.

3.7 Core wire alloy identity. The core wire alloy identity requirements shall be as specified in MIL-E-22200.

3.8 Covered electrode alloy identity. When specified (see 6.2.1), the covered electrode alloy identity requirements shall be as specified in MIL-E-22200.

4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions shall be in accordance with MIL-E-22200 except as specified herein.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Qualification tests. Qualification tests shall be as specified in MIL-E-22200 and table VI herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

TABLE VI. Summary of weld tests required for qualification.[1/]

Weld test no.	Electrode types	Diameter sizes						Welding[2/] position	[3/] X-ray	[4/] Tensile	Bend	[5/] Chemical	[6/] Brinell	Requirements	Weld[7/] joint design procedure
		5/64	3/32	1/8	5/32	3/16	1/4								
3	Group I and II	X	X	X	X	X	X	F	X	----	----	----	----	MIL-E-22200	[8/] Figure 1 and notes thereto
									----	X	----	----	X	Table II herein	
									----	----	----	X	----	Table IV herein	
									----	----	X	----	----	Table III herein	
8a	Group I	X	X	X	-	-	-	V	X	----	----	----	----	MIL-E-22200	Figure 2 and notes thereto
		-	-	-	X	X	X	F	X	----	----	----	----		

1/ Tests shall be conducted in both as-welded and stress relieved conditions for type 9N10 for qualification only.
2/ F = Flat, V = Vertical.
3/ When both tests (3 and 8a) are applicable, radiography of test no. 3 will satisfy the requirements for test no. 8a for group I, class 2 electrodes, and accordingly test no. 8a may be eliminated.
4/ Includes tensile strength and elongation. Yield strength and reduction in area shall be reported.
5/ May be determined from milling sample taken from ruptured ends of tensile specimens or weld drillings. In event of variances with composition requirements or conflicting results of testing laboratories, chemical composition shall be verified in accordance with MIL-E-22200.
6/ Brinell hardness required for group II electrodes only.
7/ Dc reverse polarity shall be used for all weld tests, with amperages and welding techniques recommended by the manufacturer. Electrical conditions shall be recorded.
8/ See figure 1 notes for number and type of test specimens.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

4.3 Comparison inspection. In addition to the examinations and tests specified in MIL-E-22200, tests specified in table VII herein shall be conducted in accordance with MIL-E-22200, when required by NAVSEA.

TABLE VII. Comparison inspection tests.

Electrode type	Sizes	Tests	Requirements
All	All	Weld test no. 3 (see table VI herein)	Tables II and III herein
All	All	MIL-E-22200	Table IV herein

4.4 Quality conformance inspection.

4.4.1 Lot. A lot of electrodes shall be as specified in MIL-E-22200 for inspection level A, except the controlled core wire chemical composition system specified in MIL-E-22200 shall not be used.

4.4.2 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be in accordance with MIL-E-22200.

4.4.3 Quality conformance inspection. Weld tests specified in table VIII shall be conducted on electrode samples selected in accordance with 4.4.2. Tests shall be conducted in the as-welded condition except as specified in table VIII (see 6.2.2).

TABLE VIII. Quality conformance tests.

Electrode types	Required tests			
	Chemical	Weld test no. 3	Weld test no. 8a[1/]	Procedures and requirements
MIL-1N12	X	X[2/]	X	Chemical
MIL-3N12	X	X[2/]	X	MIL-E-22200, table IV,
MIL-3N12H	X	X[2/]	X	and note 5 to table VI
MIL-4N1A	X	X[2/]	X	herein.
MIL-4N1AH	X	X[2/]	X	
MIL-4N11	X	X[2/]	X	
MIL-8N12	X	X[2/]	X	Weld tests
MIL-8N12H	X	X[2/]	X	
MIL-9N10	X	X[2/]	X	See table VI herein
MIL-3N1B	X	-	-	
MIL-3N1C	X	-	-	
MIL-3N1L	X	-	-	
MIL-3N1N	X	-	-	
MIL-4N1W	X	-	-	
All	Concentricity			Table X herein

See footnotes at top of next page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

1/ For class 2 electrodes wherein test no. 3 is radiographed, test no. 8a may be eliminated (see note 8 of table VI).
2/ Tests shall be conducted in the as-welded condition only.

4.4.3.1 Unsatisfactory test results. If the results of the first tests are determined to be unsatisfactory, two retests shall be permitted. The result of both retests shall be satisfactory for lot acceptance.

4.5 Test procedures.

4.5.1 Weld tests. Table VI summarizes the required weld tests for qualification testing of electrodes and for testing as specified in tables VII and VIII. Tests shall be conducted in the as-welded condition unless otherwise specified in table VI. Base plate for weld tests shall be as specified in table IX.

TABLE IX. Base plate for weld tests.1/, 2/, 3/

Electrode type	Base plate
Group I	
MIL-3N12and 3N12H	Nickel-chromium-iron alloy 4/
MIL-4N1A, 4N1AH, 8N12, 8N12H and 1N12	Nickel-chromium-iron alloy 4/ or nickel-copper alloy 5/ and mild steel 6/
MIL-9N10	Nickel-copper-alloy 5/ and mild steel 6/
MIL-4N11	Nickel 7/ or nickel-copper alloy 5/ and mild steel 6/
MIL-4N12	Nickel-chromium-iron alloy 4/ or nickel-copper alloy 5/ and mild steel
Group II	
All types	Alloy similar in composition to the electrode type under test.

1/ Where combinations of base plate materials are indicated, one member shall be nickel alloy.
2/ Base plate material for 5/64 inch diameter electrodes shall be similar alloy types only.
3/ Backing strip material shall be similar in composition to the electrode being tested or of carbon steel.
4/ Nickel-chromium-iron alloy shall be in accordance with MIL-N-23228, ASTM B 166 or ASTM B 168.
5/ Nickel-copper alloy shall be in accordance with class A of QQ-N-281 or ASTM B 127.
6/ Mild steel shall be in accordance with MIL-S-22698, ASTM A 131 or ASTM A 285.
7/ Nickel shall be in accordance with ASTM B 162.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

4.5.2 Other tests. Table X summarizes tests, in addition to those specified in table VI, required for qualification testing of electrodes, comparison inspection, and quality conformance inspection, as applicable.

TABLE X. Summary of other tests.

Tests	Requirements	Procedures
Concentricity	MIL-E-22200	MIL-E-22200
Dielectric strength 1/	MIL-E-22200	MIL-E-22200
Covering, flaking and cracking 1/	MIL-E-22200	MIL-E-22200
Covering moisture 1/	3.2.1 herein	MIL-E-22200
Alloy identity	MIL-E-22200	MIL-E-22200

1/ Required for qualification inspection only.

4.6 The identification of melts, heats or lots; the general requirements for test specimens; the rejection of nonconforming lots or specimens; and the retest and resubmittal of nonconforming lots shall be in accordance with FED-STD-151.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Packaging and packing. The welding electrodes shall be packaged level A or C and packed level A, B or C in accordance with MIL-W-10430 as specified (see 6.2.1).

5.1.1 Special instructions. When electrodes are packaged in class 1a unit containers in accordance with MIL-W-10430, each end of the container shall be provided with a circular fiberboard pad or other cushioning material to restrict the end-wise movement of the electrodes. The cushioning material shall be dry and in equilibrium with the electrode covering.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

5.2 Marking. In addition to any special marking required herein or in the contract or order (see 6.2.1), interior packages and exterior shipping containers and palletized loads shall be marked in accordance with MIL-W-10430.

5.2.1 Special marking.

5.2.1.1 Interior packages and shipping containers. In addition to the lot identification as specified in MIL-E-22200, identification on interior packages shall include MIL-type, size, specification number, manufacturer's or distributor's name, date of manufacture (month/year), and manufacturer's or distributor's brand or type designation. In addition, all unit packages shall be marked or labeled with the following warning information, or equivalent, which shall be prominently displayed:

"

WARNING: Protect yourself and others. Read and understand this label.

FUMES AND GASES can be dangerous to your health. ARC RAYS can injure eyes and burn skin. ELECTRIC SHOCK can kill.

Read and understand the manufacturer's instructions and your employer's safety practices.
Keep your head out of the fumes.
Use enough general ventilation or exhaust at the arc or both to keep fumes and gases from your breathing zone, and the general area.
Wear correct eye, ear and body protection.
Do not touch live electrical parts.
See American National Standard Z49.1 "Safety in Welding and Cutting" published by the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135; OSHA Safety and Health Standards, 29 CFR 1910, are for sale by the Superintendent of Documents, U.S. Government Printing Office Washington, DC 20402.

DO NOT REMOVE THIS LABEL

"

6. NOTES

6.1 Intended use. Nickel, nickel alloy and chromium-cobalt alloy types of welding electrodes covered by this specification are used, principally, in applications involving corrosion and heat resistance as indicated in 6.1.1 through 6.1.11.

6.1.1 MIL-1N12. The applications of this type electrode are the joining of nickel-chromium-molybdenum-columbium alloy (see ASTM B 443, B 444, and B 446) to itself and to steel and the overlaying of nickel-chromium-molybdenum-columbium alloy on steel.

6.1.2 MIL-3N12 and 3N12H. The applications of this type electrode are the joining of wrought nickel-chromium-iron alloy (see MIL-N-23228) to itself and the welding of nickel-chromium-iron alloy (welding grade) castings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

6.1.3 MIL-4N1A and 4N1AH. The applications of this type electrode are the welding of materials such as carbon steel, AISI 200, 300, 400, 500 steels, nickel and nickel base alloys to themselves or to each other in wrought and cast forms.

6.1.4 MIL-4N11. The applications of this type electrode are the joining of wrought commercially pure nickel (see ASTM B 162) to steel, overlaying of nickel on steel, and welding of the clad side of nickel-clad steel.

6.1.5 MIL-8N12 and 8N12H. The applications of this type electrode are similar to those for MIL-4N1A where higher quality levels are normally required. This type is also intended for overlay welding applications.

6.1.6 MIL-9N10. The applications of this type electrode are welding wrought nickel-copper alloys (see ASTM B 127 and QQ-N-281), and cast nickel-copper alloy (see QQ-N-288, weldable grade), joining nickel-copper alloys to steel, welding of the clad side of nickel-copper alloy clad steel and overlaying steel. For these applications, this type of electrode is suitable for use in the as-welded or stress relieved condition. These applications include all of the applications of the former types MIL-3N10, MIL-4N10 and MIL-8N10.

6.1.7 MIL-3N1B. The application of this type electrode is the joining of alloys of similar composition to the electrode.

6.1.8 MIL-3N1C. The application of this type electrode is the joining of alloys of similar composition to the electrode.

6.1.9 MIL-3N1L. The application of this type electrode is the joining of alloys of similar composition to the electrode.

6.1.10 MIL-3N1N. The application of this type electrode is the joining of alloys of similar composition to the electrode.

6.1.11 MIL-4N1W. The application of this type electrode is the joining of the Hastelloys A, B and C to steel and to dissimilar nickel bearing alloys.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, part no., and size of electrode required (see 1.2 and 6.4).
(c) Whether shorter length electrodes are desired (see table V, footnote 1).
(d) Whether 12-inch lengths of MIL-8N12 and 8N12H, 1/8-inch size are desired (see table V, footnote 2).
(e) Whether lot identification is required for individual electrodes (see 3.6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

(f) Whether alloy identity testing of the core wire of each covered electrode after final marking is required (see 3.8).
(g) Selection of applicable levels of packaging and packing (see 5.1).
(h) Special marking required (see 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.4.3	Certification data for level I material	UDI-T-23191	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., Vol. I, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

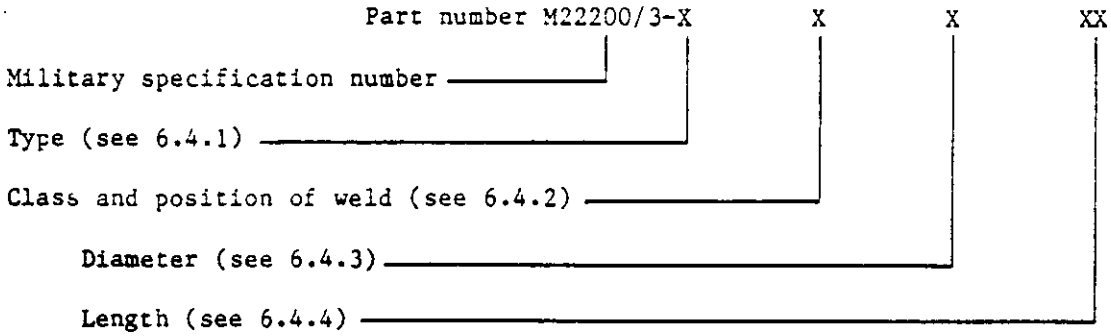
6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.3 Electrodes should be ordered by the pound.

6.3 Inspection after delivery. Post delivery inspection of electrodes to determine conformity to this specification and for acceptance thereof is the responsibility of the consignee.

6.4 Part number. A definitive part numbering system will be formulated to identify each item covered by this specification. The part number will be formulated by selecting from the required options available in this specification as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.



6.4.1 Type designation. A single letter as shown in table XI will be used to designate the required type (see table I).

TABLE XI. Type designation.

Designator	Type
A	MIL-1N12 (ENiCrM-3)
B	MIL-3N12 (ENiCrFe-1)
M	MIL-3N12H
C	MIL-4N1A (ENiCrFe-)
N	MIL-4N1AH
D	MIL-4N11 (ENi-1)
F	MIL-8N12 (ENiCrFe-3)
O	MIL-8N12H
G	MIL-9N10 (ENiCu-7)
H	MIL-3N1B (ENiMo-1)
I	MIL-3N1C (ENiMo-2)
J	MIL-3N1L
K	MIL-3N1N
L	MIL-4N1W (ENiMo-3)

6.4.2 Class designation. The numbers "1" or "2" will be used to designate the required class and welding position (see table I).

6.4.3 Diameter designation. A single letter as shown in table XII will be used to designate the required rod or wire diameter (see table I).

TABLE XII. Diameter designation.

Designator	Diameter (inch)
A	5/64
B	3/32
C	1/8
D	5/32
E	3/16
F	1/4

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

6.4.4 Length designation. The length will be designated by the numbers "9", "12", or "14" which indicate, in inches, the required rod or wire length (see table V).

6.5 Subject term (key word) listing.

Alloy
Copper-nickel alloy
Core wire alloy identity
Electrodes
Steel
Weld metal

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

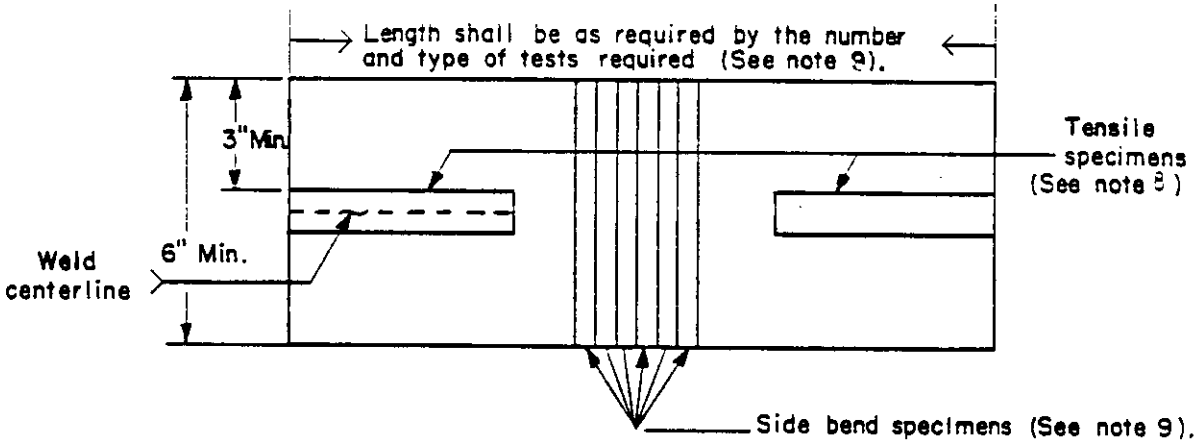
Custodians:
Army - MR
Navy - SH
Air Force - 20

Review activities:
DLA - GS

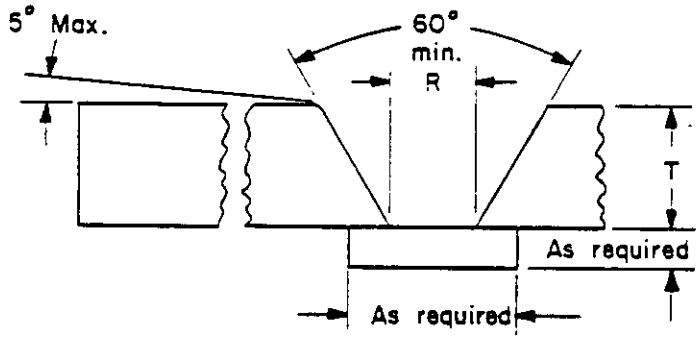
User activities:
Navy - MC, OS, YD

Preparing activity:
Navy - SH
(Project 3439-0582)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.



Layout of test plate



SH 9307A

Groove preparation of test plate

Electrode size (inch)	Minimum thickness (T) (inch)	Maximum root opening (R) (inch)	Number of layers	
			Minimum	Maximum
5/64	5/16	3/16	1/	1/
3/32	1/2	1/4	1/	1/
1/8	1/2	1/4	1/	1/
5/32	3/4	1/2	6	9
3/16	3/4	1/2	6	9
1/4	3/4	1/2	6	9

1/ Pass and layer sequence shall be recorded and reported.

FIGURE 1. Welded joint for mechanical and bend tests (test no. 3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

Notes to figure 1:

1. Weld test plates for all weld metal tensile and bend specimens shall be prepared using the plate material and electrode types as shown in table IX (see AWS B4.0 for test specimen details.)
2. Tests shall be conducted for the as-welded condition or both the as-welded and stress relieved condition as specified by table VI.
3. Welding shall be performed with plate in the flat position, using the current values and welding techniques recommended by the manufacturer.
4. Preheating shall not be used. The interpass temperature shall not exceed 300 degrees Fahrenheit (°F).
5. Welds shall be deposited in approximately 1/8-inch thick layers using a three-core wire diameter maximum weave build-up sequence. Final layers shall include a weld reinforcement of 1/32 to 1/8 inch. The root weld layer may be deposited with 3/32- or 1/8-inch diameter electrodes when testing 5/32-inch and larger diameter electrodes.
6. When stress relieved tensile and bend tests are required, stress relieving shall be accomplished by heating to 1150 ± 25°F holding at temperature for 1 hour per inch of thickness or fraction thereof and furnace cooling to a temperature below 500°F at a rate not to exceed 200°F per hour.
7. The number and type of mechanical tests shall be as follows:

Electrode type	Electrode diameter (inch)	Number and type of test specimens 1/								
		Tensile						Side	Bend	Hardness
		0.160		0.252		0.505				
		AW	HT	AW	HT	AW	HT	AW	HT	
3/ 9N10	5/64	1	1	–	–	–	–	2/ 2	2/ 2	
	3/32 and 1/8	–	–	1	1	–	–	3	3	
	All other sizes	–	–	–	–	1	1	3	3	
All other group I types	5/64	1	–	–	–	–	–	2/ 2	–	
	3/32 and 1/8	–	–	1	–	–	–	3	–	
	All other sizes	–	–	–	–	1	–	3	–	
All group II types	5/64	1	–	–	–	–	–	–	–	4/
	3/32 and 1/8	–	–	1	–	–	–	–	–	4/
	All other sizes	–	–	–	–	1	–	–	–	4/

1/ AW = as-welded, HT = heat treated.
2/ For 5/64-inch diameter electrode conduct transverse face bends.
3/ Stress relief heat treatment shall be accomplished in accordance with note 6 of figure 1. Testing of heat treated specimens is required for qualification testing only.
4/ Take a minimum of three hardness readings in accordance with MIL-E-22200 and average.

FIGURE 1. Welded joint for mechanical and bend tests (test no. 3). - Continued

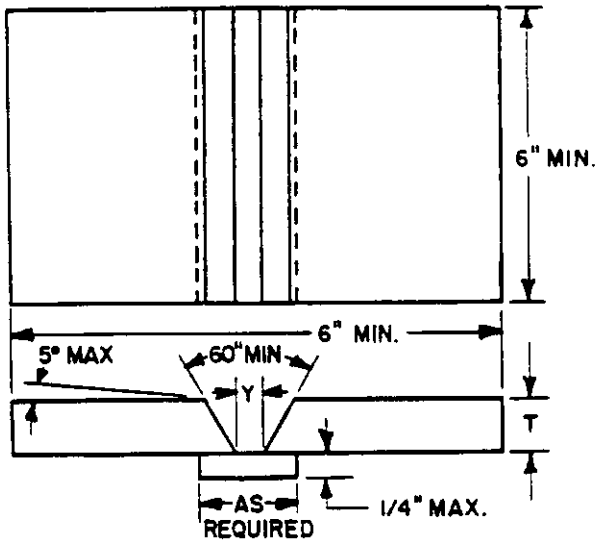
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

Notes to figure 1: - Continued

8. The results of the tensile and hardness tests shall conform to table II. The results of the bend test shall conform to table III.
9. Radiographic examination of test no. 3 is not required but may be used in lieu of radiography of test no. 8a for group 1, class 2 electrodes. When test no. 3 is employed for radiography, after completion of welding, the weld reinforcement and backing strip shall be removed flush with base plate surfaces. The assembly shall be radiographed in accordance with MIL-STD-271. Radiographs shall meet the acceptance standard of NAVSEA 0900-LP-003-9000 as specified in MIL-E-22200.

FIGURE 1. Welded joint for mechanical and bend tests (test no. 3). - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.



SH 9308B

Electrode size (inch)	5/64	3/32	1/8	5/32	3/16 and 1/4
Thickness (T) minimum (inch)	1/8	1/4	3/8	1/2	1/2
Root opening (Y) (inch)1/	1/8	1/4	5/16	3/8	1/2

1/ Tolerance plus or minus 1/16 inch.

NOTES:

1. Welding shall be accomplished in the vertical or flat position as specified in table VI.
2. Base metal shall be in accordance with table IX.
3. The bevel edge preparation shall be machined or ground.
4. Welds shall be deposited in approximately 1/8-inch thick layers. Each weld bead shall contain a start in the area to be evaluated. Weld deposits shall be made using a three-core wire diameter maximum weld weave. The root weld layer may be deposited with 3/32- or 1/8-inch diameter electrodes when testing 5/32-inch and larger diameter electrodes.
5. The completed weld reinforcement height shall be 3/16 inch maximum.
6. Preheating shall not be used.
7. After completion of the weld, the weld reinforcement and backing strip shall be removed flush with base plate surfaces and the assembly shall be X-rayed in accordance with MIL-STD-271.
8. The radiographs shall meet the acceptance standards of NAVSEA 0900-LP-003-9000 as specified in MIL-E-22200.

FIGURE 2. Groove weld usability test (test no. 8a).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-22200/3F	ELECTRODES, WELDING, COVERED: NICKEL BASE ALLOY; AND COBALT BASE ALLOY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:22Z
Check the source to verify that this is the current version before use.